Binary Distillation

Group 4
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Introduction

- The main purpose of this lab was to study the separation process of a binary mixture (EtOH and H2O)
- Separation method used was batch distillation
- Performed distillation with both total reflux and constant reflux ratios
- This lab also served to study the accuracy of the McCabe-Thiele method for analyzing a binary distillation

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Theory

- McCabe-Thiele Method Assumptions:
  - Each stage is at equilibrium
  - 100% efficiency of the stages
- Operating equations

$$\text{Top: } y = \frac{L}{V}x + \left(1 - \frac{L}{V}\right)x_D$$

$$\text{Bottom: } y = \frac{\bar{L}}{\bar{V}}x - \left(\frac{\bar{L}}{\bar{V}} - 1\right)x_B$$

- Total reflux L = V
  - Calculate efficiency of column from:

$$n + 1 = \frac{\log\left[\left(\frac{x_{EtOH}}{x_{H2O}}\right)_d \left(\frac{x_{H2O}}{x_{EtOH}}\right)_b\right]}{\log(\alpha_{EtOH/H2O})_{avg}} \quad \text{AND} \quad E = \frac{\text{number of theoretical plates}}{\text{number of actual plates}} * 100\%$$

- Constant reflux ratios
  - Use McCabe-Thiele analysis to determine efficiency

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Apparatus

- Main Equipment
  1. Column
  2. Reboiler
  3. Condenser
  4. Trays
  5. Manometer
  6. Heater
  7. Valves*
  8. Heat exchanger
  9. Reflux valve
  10. Decanter
  11. Top product receiver
  12. Bottom product receiver
  13. Cooling water flowmeter

* Note there are many valves in this experiment

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Slide 1

DL1 98/100
Donglin Li, 11/2/2019

Slide 2

DL2 Font sizes could be bigger.
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DL3 -1
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Slide 3

DL4 Type instead of pictures.
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Material & Supplies

- Material list
  - Ethanol
  - Water
  - Large graduated cylinder
  - Measuring cylinder
  - Timer
  - Bottle
  - Refractometer
  - Test tubes
  - Computer
  - Armfield console
  - Pipettes
  - Tap water
  - Water bath and rack
  - Funnel

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Procedure

- Prelab Steps
  1. Familiarize team with equipment
  2. Calculate volume for EtOH and H2O for 10 wt% solution
- Lab Day Steps
  1. Make 10 wt% EtOH and H2O solution
  2. Load in boiler and heat until constant temperatures on computer
  3. Create calibration curve for refractometer with 10 wt% increment composition mixtures of EtOH and H2O

  Part A
  1. Set column to total reflux and collect top and bottom samples for refractometer indexing, as well as taking boil-up rate and pressure drop of column at various boiler power settings

  Part B
  1. Set column to constant reflux ratio and repeat Part A step 1 (no need to take pressure drop)
  2. Repeat for various reflux ratios

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Experimental Challenges

- Difficulties
  - Understanding the operation of the column
- Things to watch out for during the experiment
  - Not to open valves too fast and overflow test tubes while collecting
  - Make sure the cooling water is on and at 3000 cc/min
  - Turn valves the correct direction
- Lessons Learned
  - Best to have one person taking data, one person performing refractometer indexing, and two people operating column
  - Check calculations for 10 wt% solution of EtOH and H2O

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Safety

- Wear gloves and goggles at all times
- Boiler and samples collected are HOT – use caution and heat gloves
- Be careful not to overflow hot samples while collecting

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Slide 5

DL6 Good.
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Slide 6

DL7 Good.
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